

LÊ VĂN CHÁNH

THE MYSTERY OF

CALCULUS



Contents

Topic 1. Several Variables (Multivariable functions)	3
Topic 2. Limits and Continuity	4
Topic 3. Partial Derivatives	7
Topic 4. Tangent Planes and Linear Approximations	9
Topic 5. The Chain Rules	12
Topic 6. Implicit Differentiation	15
Topic 7. Directional Derivatives	15
Topic 8. Maximum and Minimum Values	17
Topic 9. Double Integral	23
Topic 10. Polar coordinate	30
Topic 11. Tripple Integral	33
Topic 12. Line Integrals (Type 1+2)	38
Topic 13. Green Theorem	39
Topic 14. Applications of Line Integrals (Type 1+2)	40
Topic 15. Surface Integral	42
0.1 TRUE-FALSE QUIZ	46
Bibliography	47

 Topic 1. Several Variables (Multivariable functions)

Problem 1

Find and sketch the domain of the function.

1) $f(x, y) = \sqrt{x-2} + \sqrt{y-1}$.

6) $g(x, y) = \frac{\ln(2-x)}{1-x^2-y^2}$.

2) $f(x, y) = \sqrt[4]{x-3y}$.

7) $f(x, y) = \frac{\sqrt{y-x^2}}{1-x^2}$.

3) $f(x, y) = \ln(9-x^2-9y^2)$.

8) $f(x, y) = \sin^{-1}(x+y)$.

4) $f(x, y) = \sqrt{x^2+y^2-4}$.

9) $f(x, y, z) = \sqrt{4-x^2} + \sqrt{9-y^2} + \sqrt{1-z^2}$.

5) $g(x, y) = \frac{x-y}{x+y}$.

10) $f(x, y, z) = \ln(16-4x^2-4y^2-z^2)$.

Problem 2

Sketch the graph of the function.

1) $f(x, y) = y$.

6) $f(x, y) = 2 - x^2 - y^2$.

2) $f(x, y) = x^2$.

7) $f(x, y) = x^2 + 4y^2 + 1$.

3) $f(x, y) = 10 - 4x - 5y$.

8) $f(x, y) = \sqrt{4-x^2-y^2}$.

4) $f(x, y) = \sqrt{x^2+y^2}$.

9) $f(x, y) = \sqrt{4x^2+y^2}$.

5) $f(x, y) = x^2 + y^2$.

10) $f(x, y) = \sqrt{4-4x^2-y^2}$.

📁 Topic 2. Limits and Continuity

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11) The function is defined only when ...

- The function is defined whenever ...
- The function is defined for all $(x, y) \neq (0, 0)$.
- $(f \circ g)(x)$ is defined whenever both $g(x)$ and $f(g(x))$ are defined.
- This expression is defined when both $2 - x \geq 0$ and $2 - \sqrt{2 - x} \geq 0$.

Problem 3

Find the limit, if it exists, or show that the limit does not exist.

a) $\lim_{(x,y) \rightarrow (3,2)} (x^2 y^3 - 4y^2).$

b) $\lim_{(x,y) \rightarrow (2,-1)} \frac{x^2 y + x y^2}{x^2 - y^2}.$

c) $\lim_{(x,y) \rightarrow (3,2)} e^{\sqrt{2x-y}}.$

d) $\lim_{(x,y) \rightarrow (0,0)} \frac{5y^4 \cos^2 x}{x^4 + y^4}.$

e) $\lim_{(x,y) \rightarrow (1,0)} \frac{xy - y}{(x-1)^2 + y^2}.$

f) $\lim_{(x,y) \rightarrow (0,0)} \frac{x^3 - y^3}{x^2 + xy + y^2}.$

g) $\lim_{(x,y) \rightarrow (0,0)} \frac{xy^4}{x^4 + y^4}.$

h) $\lim_{(x,y) \rightarrow (\pi, \pi/2)} y \sin(x - y).$

i) $\lim_{(x,y) \rightarrow (0,0)} \frac{x^4 - 4y^2}{x^2 + 2y^2}.$

j) $\lim_{(x,y) \rightarrow (0,0)} \frac{y^2 \sin^2 x}{x^4 + y^4}.$

k) $\lim_{(x,y) \rightarrow (0,0)} \frac{xy}{\sqrt{x^2 + y^2}}.$

l) $\lim_{(x,y) \rightarrow (0,0)} \frac{xy^2 \cos y}{x^2 + y^4}.$

m) $\lim_{(x,y) \rightarrow (0,0)} \frac{x^2 + y^2}{\sqrt{x^2 + y^2 + 1} - 1}.$

n) $\lim_{(x,y) \rightarrow (0,0)} \frac{xy^4}{x^2 + y^8}.$

o) $\lim_{(x,y,z) \rightarrow (\pi, 0, 1/3)} e^{y^2} \tan(xz).$

p) $\lim_{(x,y,z) \rightarrow (0,0,0)} \frac{xy + yz^2 + xz^2}{x^2 + y^2 + z^4}.$

q) $\lim_{(x,y,z) \rightarrow (0,0,0)} \frac{x^2 y^2 z^2}{x^2 + y^2 + z^2}.$

r) $\lim_{x,y \rightarrow (0,0)} \frac{x^3 + y^3}{x^2 + y^2}.$

s) $\lim_{x,y \rightarrow (0,0)} (x^2 + y^2) \ln(x^2 + y^2).$

t) $\lim_{x,y \rightarrow (0,0)} \frac{e^{-x^2-y^2} - 1}{x^2 + y^2}.$

Problem 4

Determine whether the given function is continuous or discontinuous at $(0, 0)$. Explain your answer.

$$(a) f(x, y) = \begin{cases} \frac{3xy^2}{x^2 + y^2} & \text{if } (x, y) \neq (0, 0), \\ 0 & \text{if } (x, y) = (0, 0). \end{cases} \quad (e) f(x, y) = \begin{cases} \frac{xy^2}{x^2 + y^2} & \text{if } (x, y) \neq (0, 0), \\ 0 & \text{if } (x, y) = (0, 0). \end{cases}$$

$$(b) g(x, y) = \begin{cases} \frac{3xy^2}{x^2 + y^2} & \text{if } (x, y) \neq (0, 0), \\ 1 & \text{if } (x, y) = (0, 0). \end{cases} \quad (f) f(x, y) = \begin{cases} \frac{x^2 \sin y}{x^2 + y^2} & \text{if } (x, y) \neq (0, 0), \\ 0 & \text{if } (x, y) = (0, 0). \end{cases}$$

$$(c) h(x, y) = \begin{cases} \frac{xy}{x^2 + y^2} & \text{if } (x, y) \neq (0, 0), \\ 0 & \text{if } (x, y) = (0, 0). \end{cases} \quad (g) f(x, y) = \begin{cases} \frac{xy^3}{x^2 + y^4} & \text{if } (x, y) \neq (0, 0), \\ 1 & \text{if } (x, y) = (0, 0). \end{cases}$$

$$(d) k(x, y) = \begin{cases} \frac{\sin(x^2 + y^2)}{x^2 + y^2} & \text{if } (x, y) \neq (0, 0), \\ \frac{1}{2} & \text{if } (x, y) = (0, 0). \end{cases} \quad (h) f(x, y) = \begin{cases} \frac{xye^y}{x^2 + y^2} & \text{if } (x, y) \neq (0, 0), \\ 0 & \text{if } (x, y) = (0, 0). \end{cases}$$

Problem 5

Determine whether the given function is continuous on $\mathbb{R}^2$. Explain your answer.

$$(a) f(x, y) = \begin{cases} \frac{3xy^2}{x^2 + y^2} & \text{if } (x, y) \neq (0, 0), \\ 0 & \text{if } (x, y) = (0, 0). \end{cases} \quad (c) f(x, y) = \begin{cases} \frac{x^2 \sin y}{x^2 + y^2} & \text{if } (x, y) \neq (0, 0), \\ 0 & \text{if } (x, y) = (0, 0). \end{cases}$$

$$(b) f(x, y) = \begin{cases} \frac{xy^2}{x^2 + y^2} & \text{if } (x, y) \neq (0, 0), \\ 0 & \text{if } (x, y) = (0, 0). \end{cases}$$

 Topic 3. Partial Derivatives

Problem 6

Find all of the second order partial derivatives of the function.

a) $f(x, y) = x^4 + 5xy^3$.

l) $R(p, q) = \tan^{-1}(pq^2)$.

b) $f(x, y) = x^2y - 3y^4$.

m) $f(x, y) = x^y$.

c) $f(x, t) = t^2e^{-x}$.

n) $F(x, y) = \int_y^x \cos(e^t) dt$.

d) $f(x, t) = \sqrt{3x + 4t}$.

o) $F(\alpha, \beta) = \int_\alpha^\beta \sqrt{t^3 + 1} dt$.

e) $z = \ln(x + t^2)$.

p) $f(x, y, z) = x^3yz^2 + 2yz$.

f) $z = x \sin(xy)$.

q) $f(x, y, z) = xy^2e^{-xz}$.

g) $f(x, y) = \frac{x}{y}$.

r) $w = \ln(x + 2y + 3z)$.

h) $f(x, y) = \frac{x}{(x + y)^2}$.

s) $w = y \tan(x + 2z)$.

i) $f(x, y) = \frac{ax + by}{cx + dy}$.

t) $p = \sqrt{t^4 + u^2 \cos v}$.

j) $w = \frac{e^v}{u + v^2}$.

u) $u = x^{y/2}$.

k) $g(u, v) = (u^2v - v^3)^5$.

Problem 7

Find the indicated partial derivative.

a) $R(s, t) = te^{s/t}$; $R_t(0, 1)$.

b) $f(x, y) = y \sin^{-1}(xy)$; $f_y\left(1, \frac{1}{2}\right)$.

c) $f(x, y, z) = \ln\left(\frac{1 - \sqrt{x^2 + y^2 + z^2}}{1 + \sqrt{x^2 + y^2 + z^2}}\right)$; $f_y(1, 2, 2)$.

d) $f(x, y, z) = x^{yz}$; $f_z(e, 1, 0)$.

Problem 8

Prove that

a) $x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} = (x + y + \ln u)u$, where $u(x, y) = x^y y^x$.

b) $\frac{\partial f}{\partial x} + \frac{\partial f}{\partial y} = 1$ and $\frac{\partial^2 u}{\partial x^2} \cdot \frac{\partial^2 u}{\partial y^2} = \left(\frac{\partial^2 u}{\partial x \partial y} \right)^2$, where $u(x, y) = \ln(e^x + e^y)$.

c) $\frac{\partial^2 u}{\partial x^2} \cdot \frac{\partial^2 u}{\partial y^2} - u \left(\frac{\partial^2 u}{\partial x \partial y} \right)^2 = 0$, where $u(x, y) = \ln x \ln y$.

d) $2 \frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial x \partial y} = 0$, where $u(x, y) = 2 \cos^2 \left(y - \frac{x}{2} \right)$.

e) $\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = -u$, where $u(x, y) = x \sin y + y \sin x$.

f) $u_{xxx} + u_{yyyy} = xu_{xyy} + yu_{xxy}$, where $u(x, y) = xe^y + ye^x$.

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Topic 4. Tangent Planes and Linear Approximations

Problem 9

Find the equation of the tangent plane to the given surface at the specified point.

- a) $z = 3y^2 - 2x^2 + x$, $(2, -1, -3)$.
- b) $z = \sqrt{xy}$, $(1, 1, 1)$.
- c) $z = xe^{xy}$, $(2, 0, 2)$.
- d) $z = 2x^2 + y^2 - 5y$, $(1, 2, -4)$.
- e) $z = x^4 - y^2$, $(x, y) = (3, 2)$.
- f) $z = x^3 + y^2$, $(1, 2, 5)$.
- g) $z = 4x^2 - y^2 + 2y$, $(-1, 2, 4)$.
- h) $y = x^2 - z^2$, $(4, 7, 3)$.
- i) $xyz = 2$, $(1, 1, 2)$.
- j) $xyz^2 = 6$, $(3, 2, 1)$.
- k) $z = x^2y$, $(2, 1, 4)$.
- l) $z = y \ln x$, $(1, 4, 0)$.
- m) $z = (x + 2)^2 - 2(y - 1)^2 - 5$, $(2, 3, 3)$.
- n) $z = e^{x-y}$, $(2, 2, 1)$.
- o) $z = x \sin(x + y)$, $(-1, 1, 0)$.
- p) $z = \ln(x - 2y)$, $(3, 1, 0)$.
- q) $z = -9x^2 - 3y^2$, $P(2, 1, -39)$.
- r) $x^2 + 10xyz + y^2 + 8z^2 = 0$, $P(-1, -1, -1)$.
- s) $z = \ln(10x^2 + 2y^2 + 1)$, $P(0, 0, 0)$.
- t) $z = e^{7x^2 + 4y^2}$, $P(0, 0, 1)$.
- u) $xy + yz + zx = 11$, $P(1, 2, 3)$.

v) $x^2 + 4y^2 = z^2, P(3, 2, 5).$

w) $x^3 + y^3 = 3xyz, P\left(1, 2, \frac{3}{2}\right).$

x) $z = \ln \sqrt{x^2 + y^2}, P(3, 4).$

y) $z = x^2 - 2xy + y^2, P(1, 2, 1).$

Tangent Plane

The equation of the tangent plane to the surface $z = f(x, y)$ at $(x_0, y_0, f(x_0, y_0))$ is given by

$$z = f(x_0, y_0) + f_x(x_0, y_0)(x - x_0) + f_y(x_0, y_0)(y - y_0).$$

In the general case, the surface (S) is defined by the equation

$$F(x, y, z) = 0.$$

The normal to the tangent plane to (S) at $M_0(x_0, y_0, z_0)$ on (S) is

$$N = \nabla F(M_0) = (F_x(M_0), F_y(M_0), F_z(M_0)).$$

The equation of Tangent plane is then,

$$a(x - x_0) + b(y - y_0) + c(z - z_0) = 0,$$

where $N = (a, b, c)$ is the said normal vector.

Problem 10 [ABD16]

Find a point on the surface $z = 3x^2 - y^2$ at which the tangent plane is parallel to the plane $6x + 4y - z = 5$.

Problem 11 [ABD16]

Determine whether the statement is true or false. Explain your answer.

- If the tangent plane to the graph of $z = f(x, y)$ at the point $(1, 1, 2)$ has equation $x - y + 2z = 4$, then $f_x(1, 1) = 1$ and $f_y(1, 1) = -1$.
- If the tangent plane to the graph of $z = f(x, y)$ at the point $(1, 2, 1)$ has equation $2x + y - z = 3$, then the local linear approximation to f at $(1, 2)$ is given by the function $L(x, y) = 1 + 2(x - 1) + (y - 2)$.

Problem 12

Explain why the function is differentiable at the given point. Then find the linearization $L(x, y)$ of the function at that point.

a) $f(x, y) = \sqrt{xy}, (1, 4).$

- b) $f(x, y) = x^2 e^y, \quad (1, 0).$
- c) $f(x, y) = \frac{1+y}{1+x}, \quad (1, 3).$
- d) $f(x, y) = 4 \arctan(xy), \quad (1, 1).$
- e) $f(x, y) = y + \sin(x/y), \quad (0, 3).$

Problem 13

Using a linear approximation to estimate.

- (a) $f(0.9, 1.01)$, where $f(x, y) = x^3 y^4$.
- (b) $f(6.9, 2.06)$, where $f(x, y) = \ln(x - 3y)$.
- (c) $f(3.2, 1.8)$, where $f(x, y) = \ln(x^2 - y^3)$.
- (d) $f(2.02, 2.98)$, where $f(x, y) = 1 + \ln(xy - 5)$.
- (e) $f(3, 0, 001)$, where $f(x, y) = \sqrt{x + e^{4y}}$.
- (f) $f(2.01, 0.99)$, where $f(x, y) = \frac{x}{x + y}$.

Problem 14

Given that f is a differentiable function with $f(2, 5) = 6$, $f_x(2, 5) = 1$, and $f_y(2, 5) = -1$, use a linear approximation to estimate $f(2.2, 4.9)$.

Problem 15

Using a linear approximation to estimate

- (a) $A = \sqrt{1.02^3 + 1.97^3}$.
- (b) $B = \ln\left(\sqrt[3]{1.03} + \sqrt[4]{0.981}\right)$, $\ln 2 \simeq 0.69$.
- (c) $\ln(\sqrt{1.04} + \sqrt[4]{0.96} - 1)$.
- (d) $\sqrt[3]{(1.02)^2 + (0.05)^2}$.
- (e) $\sqrt{(4.05)^2 + (3.07)^2}$.
- (f) $\sqrt[3]{(2.06)^2 + (1.97)^2}$.
- (g) $(2.01)^{3.03}$, $\ln 2 \simeq 0.69$.
- (h) $(1.03)^{3.01}$.
- (i) $(1.95)^2 e^{0.02}$.
- (j) $0.98 \ln(1.01)$.
- (k) $\arctan\left(\frac{0.02}{1.99}\right)$.
- (l) $\sin(29^\circ) \cos 72^\circ$.

Problem 16 [ABD16]

A function f is given along with a local linear approximation L to f at a point P . Use the information given to determine point P .

- a) $f(x, y) = x^2 + y^2; L(x, y) = 2y - 2x - 2$.
- b) $f(x, y) = x^2y; L(x, y) = 4y - 4x + 8$.
- c) $f(x, y, z) = xy + z^2; L(x, y, z) = y + 2z - 1$.
- d) $f(x, y, z) = xyz; L(x, y, z) = x - y - z - 2$.

Problem 17

Verify the linear approximation at $(0, 0)$.

- a) $\frac{2x+3}{4y+1} \approx 3 + 2x - 12y$.
- b) $\sqrt{y + \cos^2 x} \approx 1 + x + \frac{1}{2}y$.
- c) $e^x \cos(xy) \approx x + 1$.
- d) $\frac{y-1}{x+1} \approx x + y - 1$.

Linear approximation

The linear approximation $L(x, y)$ to f at (a, b) : $f(x, y) \approx L(x, y) = f(a, b) + (x - a)f_x(a, b) + (y - b)f_y(a, b)$.

 Topic 5. The Chain Rules

Problem 18

Find $\frac{\partial F}{\partial s}$ and $\frac{\partial F}{\partial t}$ where $F(s, t) = f(x(s, t), y(s, t))$.

- a) $f(x, y) = x^2y^3, x(s, t) = s \cos t, y(s, t) = s \sin t$.
- b) $f(x, y) = \arcsin(x - y), x = s^2 + t^2, y = 1 - 2st$.
- c) $f(x, y) = e^{x+2y}, x(s, t) = \frac{s}{t}, y(s, t) = \frac{t}{s}$.
- d) $f(x, y) = \tan\left(\frac{x}{y}\right), x = 2s + 3t, y = 3s - 2t$.

Problem 19

Find $\frac{\partial F}{\partial s}$ and $\frac{\partial F}{\partial t}$ where $F(s, t) = f(r(s, t), \theta(s, t))$.

a) $f(r, \theta) = \sin \theta \cos r, r = st^2, \theta = s^2t$.

b) $f(r, \theta) = e^r \cos \theta, r = st, \theta = \sqrt{s^2 + t^2}$.

Problem 20

Let f be differentiable on $\mathbb{R}^2$, $x(s, t) = s + t, y(s, t) = s - t, F(s, t) = f(x(s, t), y(s, t))$. Prove that

$$\left(\frac{\partial F}{\partial s}\right)^2 + \left(\frac{\partial F}{\partial t}\right)^2 = 2\left(\frac{\partial f}{\partial x}\right)^2 + 2\left(\frac{\partial f}{\partial y}\right)^2.$$

Problem 21

Let f be differentiable on $\mathbb{R}^2$, $x(s, t) = s + t, y(s, t) = s - t, F(s, t) = f(x(s, t), y(s, t))$. Prove that

$$\left(\frac{\partial f}{\partial x}\right)^2 - \left(\frac{\partial f}{\partial y}\right)^2 = \frac{\partial F}{\partial s} \frac{\partial F}{\partial t}.$$

Problem 22

Suppose f is a differentiable function of two variables. The following table gives the values of f, f_x , and f_y at selected points:

(x, y)	$f(x, y)$	$f_x(x, y)$	$f_y(x, y)$
$(0, 0)$	3	4	8
$(1, 2)$	6	2	5

Use the chain rule to calculate the following derivatives:

a) Let

$$g(u, v) = f(e^u + \sin v, e^u + \cos v).$$

Compute

$$g_u(0, 0) \quad \text{and} \quad g_v(0, 0).$$

b) Let

$$g(r, s) = f(2r - s, s^2 - 4r).$$

Compute

$$g_r(1, 2) \quad \text{and} \quad g_s(1, 2).$$

Problem 23

The temperature at a point (x, y) is $T(x, y)$, measured in degrees Celsius. A bug crawls so that its position after t seconds is given by

$$x = \sqrt{1+t}, \quad y = 2 + \frac{1}{3}t,$$

where x and y are measured in centimeters. The temperature function satisfies

$$T_x(2, 3) = 4, \quad T_y(2, 3) = 3.$$

How fast is the temperature rising on the bug's path after 3 seconds?

Problem 24

Let f be differentiable on $\mathbb{R}^2$, $u(s, t) = f(x(s, t), y(s, t))$, where $x(s, t) = e^s \cos t$ and $y(s, t) = e^s \sin t$. Show that

a)

$$\left(\frac{\partial u}{\partial x}\right)^2 + \left(\frac{\partial u}{\partial y}\right)^2 = e^{-2s} \left[\left(\frac{\partial u}{\partial s}\right)^2 + \left(\frac{\partial u}{\partial t}\right)^2 \right].$$

b)

$$\frac{\partial^2 f}{\partial x^2} + \frac{\partial^2 f}{\partial y^2} = e^{-2s} \left[\frac{\partial^2 u}{\partial s^2} + \frac{\partial^2 u}{\partial t^2} \right].$$

Problem 25

Let f have continuous second-order partial derivatives on $\mathbb{R}^2$. Define

$$z = f(x, y), \quad x = 2s + 3t, \quad y = 3s - 2t.$$

Compute the mixed partial derivative

$$\frac{\partial^2 z}{\partial s \partial t}.$$

Problem 26

Let f have continuous second-order partial derivatives on $\mathbb{R}^2$. Define

$$z = f(x, y), \quad x = t \cos s, \quad y = t \sin s.$$

Compute the mixed partial derivative

$$\frac{\partial^2 z}{\partial s \partial t}.$$

Topic 6. Implicit Differentiation

Topic 7. Directional Derivatives

Problem 27

Find the directional derivative of f at the given point in the direction indicated by the angle θ .

- a) $f(x, y) = xy^3 - x^2$, $(1, 2)$, $\theta = \pi/3$.
- b) $f(x, y) = y \cos(xy)$, $(0, 1)$, $\theta = \pi/4$.
- c) $f(x, y) = \sqrt{2x + 3y}$, $(3, 1)$, $\theta = -\pi/6$.

Problem 28

Find the directional derivative of the function at the given point in the direction of the vector $\mathbf{v}$.

- a) $f(x, y) = e^x \sin y$, $(0, \pi/3)$, $\mathbf{v} = \langle -6, 8 \rangle$.
- b) $f(x, y) = \frac{x}{x^2 + y^2}$, $(1, 2)$, $\mathbf{v} = \langle 3, 5 \rangle$.
- c) $g(s, t) = s\sqrt{t}$, $(2, 4)$, $\mathbf{v} = 2\mathbf{i} - \mathbf{j}$.
- d) $g(u, v) = u^2 e^{-v}$, $(3, 0)$, $\mathbf{v} = 3\mathbf{i} + 4\mathbf{j}$.
- e) $f(x, y, z) = x^2 y + y^2 z$, $(1, 2, 3)$, $\mathbf{v} = \langle 2, -1, 2 \rangle$.

Problem 29

- a) Find the directional derivative of $f(x, y, z) = xy^2 z^3$ at $P(2, 1, 1)$ in the direction of $Q(0, -3, 5)$.
- b) Find the directional derivative of $f(x, y) = \sqrt{xy}$ at $P(2, 8)$ in the direction of $Q(5, 4)$.

Problem 30

Find the rate of change of f at P in the direction of the vector $\mathbf{u}$.

- a) $f(x, y) = x/y$, $P(2, 1)$, $\mathbf{u} = \frac{3}{5}\mathbf{i} + \frac{4}{5}\mathbf{j}$.
- b) $f(x, y) = x^2 \ln y$, $P(3, 1)$, $\mathbf{u} = -\frac{5}{13}\mathbf{i} + \frac{12}{13}\mathbf{j}$.
- c) $f(x, y, z) = x^2 y z - x y z^3$, $P(2, -1, 1)$, $\mathbf{u} = \left\langle 0, \frac{4}{5}, -\frac{3}{5} \right\rangle$.
- d) $f(x, y, z) = y^2 e^{xyz}$, $P(0, 1, -1)$, $\mathbf{u} = \left\langle \frac{3}{13}, \frac{4}{13}, \frac{12}{13} \right\rangle$.

Problem 31

Find the maximum rate of change of f at the given point and the direction in which it occurs.

a) $f(x, y) = 4y\sqrt{x}$, $(4, 1)$.

c) $f(x, y) = \sin(xy)$, $(1, 0)$.

b) $f(s, t) = te^{st}$, $(0, 2)$.

d) $f(x, y, z) = x \ln(yz)$, $\left(1, 2, \frac{1}{2}\right)$.

Problem 32

a) Find the direction in which the function $f(x, y) = x^4y - x^2y^3$ decreases fastest at the point $(2, -3)$.

b) Find the directions in which the directional derivative of $f(x, y) = x^2 + xy^3$ at the point $(2, 1)$ has the value 2.

c) Find all points at which the direction of fastest change of the function $f(x, y) = x^2 + y^2 - 2x - 4y$ is $\mathbf{i} + \mathbf{j}$.

Problem 33

Near a buoy, the depth of a lake at the point with coordinates (x, y) is $z = 200 + 0.02x^2 - 0.001y^3$, where x, y , and z are measured in meters. A fisherman in a small boat starts at the point $(80, 60)$ and moves toward the buoy, which is located at $(0, 0)$. Is the water under the boat getting deeper or shallower when he departs? Explain.

Problem 34

The temperature T in a metal ball is inversely proportional to the distance from the center of the ball, which we take to be the origin. The temperature at the point $(1, 2, 2)$ is 120° .

a) Find the rate of change of T at $(1, 2, 2)$ in the direction toward the point $(2, 1, 3)$.

b) Show that at any point in the ball the direction of greatest increase in temperature is given by a vector that points toward the origin.

Problem 35

The temperature at a point (x, y, z) is given by

$$T(x, y, z) = 200e^{-x^2-3y^2-9z^2}$$

where T is measured in $^\circ\text{C}$ and x, y, z in meters.

a) Find the rate of change of temperature at the point $P(2, -1, 2)$ in the direction toward the point $(3, -3, 3)$

- b) In which direction does the temperature increase fastest at P ?
- c) Find the maximum rate of increase at P .

Problem 36

Suppose that over a certain region of space the electrical potential V is given by $V(x, y, z) = 5x^2 - 3xy + xyz$.

- a) Find the rate of change of the potential at $P(3, 4, 5)$ in the direction of the vector $\mathbf{v} = \mathbf{i} + \mathbf{j} - \mathbf{k}$
- b) In which direction does V change most rapidly at P ?
- c) What is the maximum rate of change at P ?

Problem 37

Suppose you are climbing a hill whose shape is given by the equation $z = 1000 - 0.005x^2 - 0.01y^2$, where x, y , and z are measured in meters, and you are standing at a point with coordinates $(60, 40, 966)$. The positive x -axis points east and the positive y -axis points north.

- a) If you walk due south, will you start to ascend or descend? At what rate?
- b) If you walk northwest, will you start to ascend or descend? At what rate?
- c) In which direction is the slope largest? What is the rate of ascent in that direction? At what angle above the horizontal does the path in that direction begin?

Topic 8. Maximum and Minimum Values

Problem 38

Find the local maximum and minimum values and saddle point(s) of the function.

- a) $f(x, y) = x^2 + xy + y^2 + y$.
- b) $f(x, y) = xy - 2x - 2y - x^2 - y^2$.
- c) $f(x, y) = (x - y)(1 - xy)$.
- d) $f(x, y) = y(e^x - 1)$.
- e) $f(x, y) = x^2 + y^4 + 2xy$.
- f) $f(x, y) = 2 - x^4 + 2x^2 - y^2$.

g) $f(x, y) = x^3 - 3x + 3xy^2$.

h) $f(x, y) = x^3 + y^3 - 3x^2 - 3y^2 - 9x$.

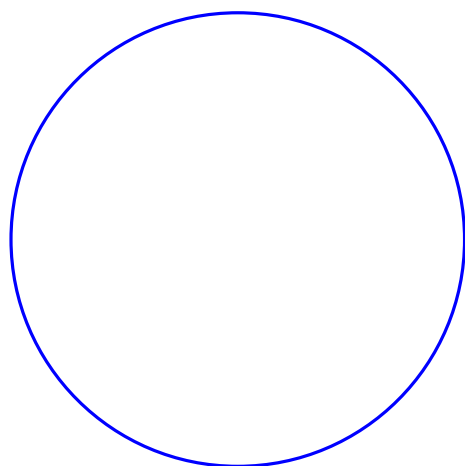
i) $f(x, y) = x^4 - 2x^2 + y^3 - 3y$.

j) $f(x, y) = xye^{-(x^2+y^2)/2}$.

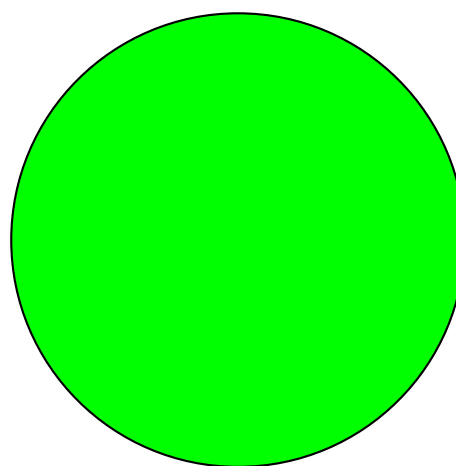
k) $f(x, y) = xy + e^{-xy}$.

l) $f(x, y) = (x^2 + y^2) e^{-x}$.

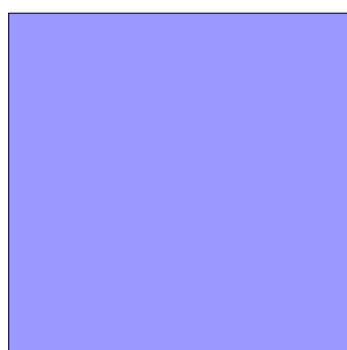
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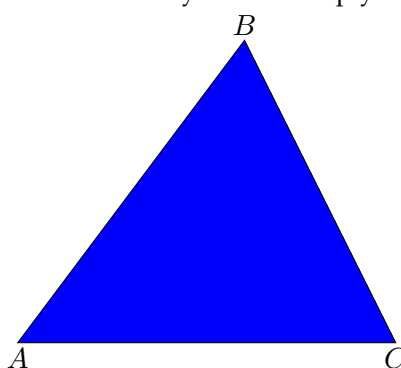
(a) Smooth boundary and empty interior



(b) Smooth boundary and nonempty interior



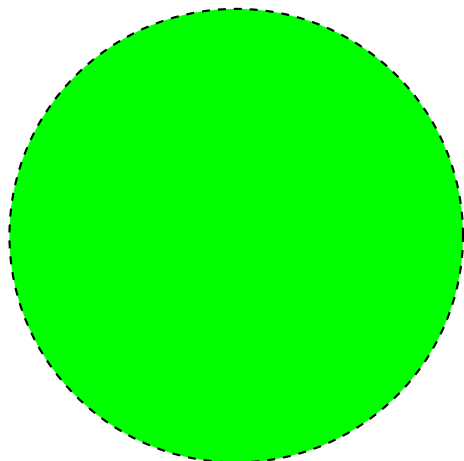
(c) Nonsmooth boundary and nonempty interior



(d) Nonsmooth boundary and nonempty interior

Figure 1: Closed and bounded domain

EXAMPLE 0.0.1.



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To find the absolute maximum and minimum values of a continuous function f on a closed, bounded set D :

- 1) Find the values of f at the critical points of f in D .
- 2) Find the extreme values of f on the boundary of D (or the value of f at candidate points).
- 3) The largest of the values from steps 1 and 2 is the absolute maximum value; the smallest of these values is the absolute minimum value.

$$\max_D f = \max \left\{ \max_S f, \max_{\partial D} f \right\},$$

$$\min_D f = \min \left\{ \min_S f, \min_{\partial D} f \right\},$$

where S is the set of stationary points of f inside of D .

If $D = E \cup F$ and f attains an absolute maximum value on E and on F then $\max_D f$ exists and $\max_D f = \max \left\{ \max_E f, \max_F f \right\}$. Similarly, we have a statement for the absolute minimum value of f .

Problem 39

Use Lagrange multipliers to find the extreme values of the function subject to the given constraint.

- a) $f(x, y) = 3x + y; \quad x^2 + y^2 = 10$.
- b) $f(x, y) = xy; \quad 4x^2 + y^2 = 8$.
- c) $f(x, y) = xe^y; \quad x^2 + y^2 = 2$.
- d) $f(x, y, z) = 2x + 2y + z; \quad x^2 + y^2 + z^2 = 9$.

Problem 40

Find the extreme values of f on the region described by the inequality.

- a) $f(x, y) = 2x^3 + y^4, \quad D = \{(x, y) \mid x^2 + y^2 \leq 1\}$.
- b) $f(x, y) = x^2 + y^2 + 4x - 4y, \quad x^2 + y^2 \leq 9$.

c) $f(x, y) = 2x^2 + 3y^2 - 4x - 5, \quad x^2 + y^2 \leq 16.$

d) $f(x, y) = e^{-xy}, \quad x^2 + 4y^2 \leq 1.$

Problem 41

Find the absolute maximum and minimum values of f on the set D .

a) $f(x, y) = x^2 + y^2 - 2x, \quad D$ is the closed triangular region with vertices $(2, 0)$, $(0, 2)$, and $(0, -2)$.

b) $f(x, y) = x + y - xy, \quad D$ is the closed triangular region with vertices $(0, 0)$, $(0, 2)$, and $(4, 0)$.

c) $f(x, y) = x^2 + y^2 + x^2y + 4, \quad D = \{(x, y) \mid |x| \leq 1, |y| \leq 1\}.$

d) $f(x, y) = x^2 + xy + y^2 - 6y, \quad D = \{(x, y) \mid -3 \leq x \leq 3, 0 \leq y \leq 5\}.$

e) $f(x, y) = x^2 + 2y^2 - 2x - 4y + 1, \quad D = \{(x, y) \mid 0 \leq x \leq 2, 0 \leq y \leq 3\}.$

f) $f(x, y) = xy^2, \quad D = \{(x, y) \mid x \geq 0, y \geq 0, x^2 + y^2 \leq 3\}.$

g) $f(x, y) = x^3 - 3x - y^3 + 12y, \quad D$ is the quadrilateral whose vertices are $(-2, 3)$, $(2, 3)$, $(2, 2)$, and $(-2, -2)$.

h) $f(x, y) = x^2y^3(1 - x - y), \quad D$ is the closed triangular region bounded by the lines: $x = 0, y = 0, x + y = 1$.

Problem 42

a) Find the shortest distance from the point $(2, 0, -3)$ to the plane $x + y + z = 1$.

b) Find the point on the plane $x - 2y + 3z = 6$ that is closest to the point $(0, 1, 1)$.

c) Find the points on the cone $z^2 = x^2 + y^2$ that are closest to the point $(4, 2, 0)$.

d) Find the points on the surface $y^2 = 9 + xz$ that are closest to the origin.

e) Find three positive numbers whose sum is 100 and whose product is a maximum.

f) Find three positive numbers whose sum is 12 and the sum of whose squares is as small as possible.

g) Find the maximum volume of a rectangular box that is inscribed in a sphere of radius r .

h) Find the dimensions of the box with volume 1000 cm^3 that has minimal surface area.

i) Find the volume of the largest rectangular box in the first octant with three faces in the coordinate planes and one vertex in the plane $x + 2y + 3z = 6$.

- j) Find the dimensions of the rectangular box with largest volume if the total surface area is given as 64 cm^2 .
- k) Find the dimensions of a rectangular box of maximum volume such that the sum of the lengths of its 12 edges is a constant c .
- l) The base of an aquarium with given volume V is made of slate and the sides are made of glass. If slate costs five times as much (per unit area) as glass, find the dimensions of the aquarium that minimize the cost of the materials.
- m) A cardboard box without a lid is to have a volume of $32,000 \text{ cm}^3$. Find the dimensions that minimize the amount of cardboard used.

Problem 43

The Shannon index (sometimes called the Shannon-Wiener index or Shannon-Weaver index) is a measure of diversity in an ecosystem. For the case of three species, it is defined as

$$H = -p_1 \ln p_1 - p_2 \ln p_2 - p_3 \ln p_3$$

where p_i is the proportion of species i in the ecosystem.

- a) Express H as a function of two variables using the fact that $p_1 + p_2 + p_3 = 1$
- b) What is the domain of H ?
- c) Find the maximum value of H . For what values of p_1, p_2, p_3 does it occur?

Topic 9. Double Integral

$$\iint_R f(x, y) dA = \lim_{m, n \rightarrow \infty} \sum_{i=1}^n \sum_{j=1}^m f(x_{ij}^*, y_{ij}^*) \Delta A.$$

$$R_{ij} \ni (x_{ij}^*, y_{ij}^*) = \left\{ \begin{array}{l} \text{the lower left corners of the rectangles} \\ \text{the lower right corners} \\ \text{the upper left corners} \\ \text{the upper right corners} \\ \text{the Midpoint Rule} \end{array} \right.$$

Problem 44

If $R = [0, 4] \times [-1, 2]$, use a Riemann sum with $m = 2, n = 3$ to estimate the value of $\iint_R (1 - xy^2) \, dA$. Take the sample points to be (a) the lower right corners and (b) the upper left corners of the rectangles.

Problem 45

- Estimate the volume of the solid that lies below the surface $z = 1 + x^2 + 3y$ and above the rectangle $R = [1, 2] \times [0, 3]$. Use a Riemann sum with $m = n = 2$ and choose the sample points to be lower left corners.
- Use the Midpoint Rule to estimate the volume in part a).

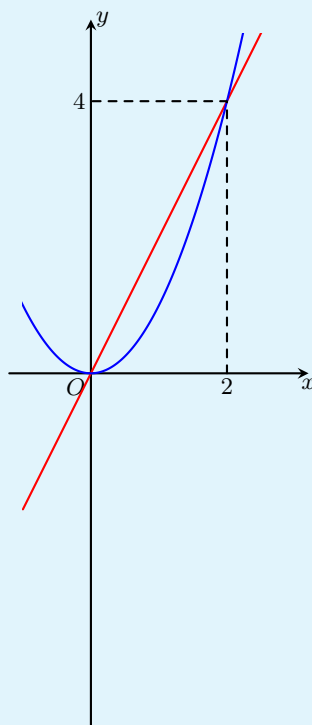
EXAMPLE 0.0.2.

- Calculate the double integral $\iint_{D_1} (6xy + 2) \, dA$ where $D_1 = \{(x, y) : 0 \leq x \leq 1, 0 \leq y \leq x^2\}$.
- Calculate the double integral. $\iint_{D_2} (6xy + 2) \, dA$ where $D_2 = \{(x, y) : \sqrt{y} \leq x \leq 1, 0 \leq y \leq 1\}$.

SOLUTION.

□

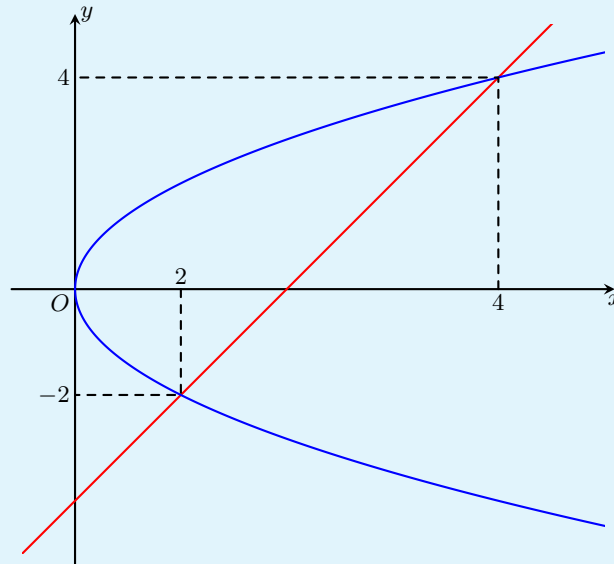
EXAMPLE 0.0.3. Calculate $\iint_D xy^2 dx dy$, where D is enclosed by the curve: $y = 2x$ and $y = x^2$.



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EXAMPLE 0.0.4. Calculate $\iint_D xy^2 dx dy$ where D is enclosed by the curve: $y = x - 4$ and $y^2 = 2x$.

SOLUTION.



□

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Problem 46

Calculate the iterated integral.

a) $\int_1^4 \int_0^2 (6x^2y - 2x) dy dx.$

j) $\int_1^3 \int_1^5 \frac{\ln y}{xy} dy dx.$

b) $\int_0^1 \int_0^1 (x + y)^2 dx dy.$

k) $\int_0^1 \int_0^2 ye^{x-y} dx dy.$

c) $\int_0^1 \int_1^2 (x + e^{-y}) dx dy.$

l) $\int_0^1 \int_0^1 \sqrt{s+t} ds dt.$

d) $\int_0^{\pi/6} \int_0^{\pi/2} (\sin x + \sin y) dy dx.$

m) $\int_1^5 \int_0^x (8x - 2y) dy dx.$

e) $\int_{-3}^3 \int_0^{\pi/2} (y + y^2 \cos x) dx dy.$

n) $\int_0^1 \int_0^y xe^{y^3} dx dy.$

f) $\int_1^4 \int_1^2 \left(\frac{x}{y} + \frac{y}{x} \right) dy dx.$

o) $\int_0^1 \int_0^{s^2} \cos(s^3) dt ds.$

g) $\int_0^3 \int_0^{\pi/2} t^2 \sin^3 \phi d\phi dt.$

p) $\int_0^2 \int_0^{y^2} x^2 y dx dy.$

h) $\int_0^1 \int_0^1 xy \sqrt{x^2 + y^2} dy dx.$

q) $\int_0^{\pi/2} \int_0^x x \sin y dy dx.$

i) $\int_0^1 \int_0^1 v(u + v^2)^4 du dv.$

r) $\int_0^1 \int_0^{e^v} \sqrt{1 + e^v} dw dv.$

Problem 47

Calculate the double integral.

a) $\iint_R (y + xy^{-2}) dA, \quad R = \{(x, y) \mid 0 \leq x \leq 2, 1 \leq y \leq 2\}.$

b) $\iint_R \frac{xy^2}{x^2 + 1} dA, \quad R = \{(x, y) \mid 0 \leq x \leq 1, -3 \leq y \leq 3\}.$

c) $\iint_R \frac{\tan \theta}{\sqrt{1 - t^2}} dA, \quad R = \left\{ (\theta, t) \mid 0 \leq \theta \leq \pi/3, 0 \leq t \leq \frac{1}{2} \right\}.$

d) $\iint_R x \sin(x + y) dA, \quad R = [0, \pi/6] \times [0, \pi/3].$

e) $\iint_R \frac{x}{1 + xy} dA, \quad R = [0, 1] \times [0, 1].$

f) $\iint_R ye^{-xy} dA, \quad R = [0, 2] \times [0, 3].$

g) $\iint_R \frac{1}{1 + x + y} dA, \quad R = [1, 3] \times [1, 2].$

- h) $\iint_D \frac{y}{x^2 + 1} dA$, $D = \{(x, y) \mid 0 \leq x \leq 4, 0 \leq y \leq \sqrt{x}\}$.
- i) $\iint_D (2x + y) dA$, $D = \{(x, y) \mid 1 \leq y \leq 2, y - 1 \leq x \leq 1\}$.
- j) $\iint_D e^{-y^2} dA$, $D = \{(x, y) \mid 0 \leq y \leq 3, 0 \leq x \leq y\}$.
- k) $\iint_D y\sqrt{x^2 - y^2} dA$, $D = \{(x, y) \mid 0 \leq x \leq 2, 0 \leq y \leq x\}$.

Problem 48

Calculate the double integral.

- a) $\iint_D x dA$, D is enclosed by the lines $y = x, y = 0, x = 1$.
- b) $\iint_D xy dA$, D is enclosed by the curves $y = x^2, y = 3x$.
- c) $\iint_D y dA$, D is bounded by $y = x - 2, x = y^2$.
- d) $\iint_D y^2 e^{xy} dA$, D is bounded by $y = x, y = 4, x = 0$.
- e) $\iint_D x \cos y dA$, D is bounded by $y = 0, y = x^2, x = 1$.
- f) $\iint_D (x^2 + 2y) dA$, D is bounded by $y = x, y = x^3, x \geq 0$.
- g) $\iint_D y^2 dA$, D is the triangular region with vertices $(0, 1), (1, 2), (4, 1)$.
- h) $\iint_D xy dA$, D is enclosed by the quarter-circle $y = \sqrt{1 - x^2}, x \geq 0$, and the axes
- i) $\iint_D (2x - y) dA$, D is bounded by the circle with center the origin and radius 2.
- j) $\iint_D y dA$, D is the triangular region with vertices $(0, 0), (1, 1)$, and $(4, 0)$.

Problem 49

Evaluate the integral.

- | | |
|--|--|
| <p>a) $\int_0^1 \int_{3y}^3 e^{x^2} dx dy.$</p> <p>b) $\int_0^1 \int_{x^2}^1 \sqrt{y} \sin y dy dx.$</p> <p>c) $\int_0^1 \int_{\sqrt{x}}^1 \sqrt{y^3 + 1} dy dx.$</p> | <p>d) $\int_0^2 \int_{y/2}^1 y \cos(x^3 - 1) dx dy.$</p> <p>e) $\int_0^1 \int_{\arcsin y}^{\pi/2} \cos x \sqrt{1 + \cos^2 x} dx dy.$</p> <p>f) $\int_0^8 \int_{\sqrt[3]{y}}^2 e^{x^4} dx dy.$</p> |
|--|--|

Problem 50

- a) Find the volume of the solid that lies under the plane $4x + 6y - 2z + 15 = 0$ and above the rectangle $R = \{(x, y) \mid -1 \leq x \leq 2, -1 \leq y \leq 1\}$.
- b) Find the volume of the solid that lies under the hyperbolic paraboloid $z = 3y^2 - x^2 + 2$ and above the rectangle $R = [-1, 1] \times [1, 2]$.
- c) Find the volume of the solid lying under the elliptic paraboloid $\frac{x^2}{4} + \frac{y^2}{9} + z = 1$ and above the rectangle $R = [-1, 1] \times [-2, 2]$.
- d) Find the volume of the solid enclosed by the surface $z = x^2 + xy^2$ and the planes $z = 0, x = 0, x = 5$ and $y = \pm 2$.
- e) Find the volume of the solid enclosed by the surface $z = 1 + x^2 ye^y$ and the planes $z = 0, x = \pm 1, y = 0$ and $y = 1$.
- f) Find the volume of the solid in the first octant bounded by the cylinder $z = 16 - x^2$ and the plane $y = 5$.
- g) Find the volume of the solid enclosed by the paraboloid $z = 2 + x^2 + (y - 2)^2$ and the planes $z = 1, x = 1, x = -1, y = 0$, and $y = 4$.
- h) Under the plane $3x + 2y - z = 0$ and above the region enclosed by the parabolas $y = x^2$ and $x = y^2$.
- i) Under the surface $z = 1 + x^2 y^2$ and above the region enclosed by $x = y^2$ and $x = 4$.
- j) Under the surface $z = xy$ and above the triangle with vertices $(1, 1), (4, 1)$, and $(1, 2)$.
- k) Enclosed by the paraboloid $z = x^2 + y^2 + 1$ and the planes $x = 0, y = 0, z = 0$, and $x + y = 2$.
- l) The tetrahedron enclosed by the coordinate planes and the plane $2x + y + z = 4$.
- m) Bounded by the planes $z = x, y = x, x + y = 2$, and $z = 0$.

- n) Enclosed by the cylinders $z = x^2, y = x^2$ and the planes $z = 0, y = 4$.
- o) Bounded by the cylinder $y^2 + z^2 = 4$ and the planes $x = 2y, x = 0, z = 0$ in the first octant.
- p) Bounded by the cylinder $x^2 + y^2 = 1$ and the planes $y = z, x = 0, z = 0$ in the first octant.

□ Topic 10. Polar coordinate

Problem 51

Sketch the region whose area is given by the integral and evaluate the integral.

- a) $\iint_{[-1,1] \times [0,2]} xy dA.$
- b) $\int_0^{\pi/6} \int_0^{\cos \theta} r dr d\theta.$
- c) $\int_{\pi/4}^{3\pi/4} \int_1^2 r dr d\theta.$
- d) $\int_{\pi/2}^{\pi} \int_0^{2 \sin \theta} r dr d\theta.$

Problem 52

Evaluate the given integral by changing to polar coordinates.

- a) $\iint_D x^2 y dA$, where D is the top half of the disk with center the origin and radius 5.
- b) $\iint_R (2x - y) dA$, where R is the region in the first quadrant enclosed by the circle $x^2 + y^2 = 4$ and the lines $x = 0$ and $y = x$.
- c) $\iint_R \sin(x^2 + y^2) dA$, where R is the region in the first quadrant between the circles with center the origin and radii 1 and 3.
- d) $\iint_R \frac{y^2}{x^2 + y^2} dA$, where R is the region that lies between the circles $x^2 + y^2 = a^2$ and $x^2 + y^2 = b^2$ with $0 < a < b$.
- e) $\iint_D e^{-x^2 - y^2} dA$, where D is the region bounded by the semicircle $x = \sqrt{4 - y^2}$ and the y -axis.
- f) $\iint_D \cos \sqrt{x^2 + y^2} dA$, where D is the disk with center the origin and radius 2.
- g) $\iint_R \arctan(y/x) dA$, where $R = \{(x, y) \mid 1 \leq x^2 + y^2 \leq 4, 0 \leq y \leq x\}$.
- h) $\iint_D e^{(x^2 + y^2)^2} dA$, where D is the disk with center the origin and radius 1.

- i) $\iint_D xy\sqrt{1+x^2+y^2}dA$, where D is the portion of the disk $x^2 + y^2 \leq 1$ that lies in the first quadrant.
- j) $\iint_D x dA$, where D is the region in the first quadrant that lies between the circles $x^2 + y^2 = 4$ and $x^2 + y^2 = 2x$.

Problem 53

Use a double integral to find the area of the region.

- One loop of the rose $r = \cos 3\theta$.
- The region enclosed by both of the cardioids $r = 1 + \cos \theta$ and $r = 1 - \cos \theta$.
- The region inside the circle $(x - 1)^2 + y^2 = 1$ and outside the circle $x^2 + y^2 = 1$.
- The region inside the cardioid $r = 1 + \cos \theta$ and outside the circle $r = 3 \cos \theta$.

Problem 54

Use polar coordinates to find the volume of the given solid.

- Under the paraboloid $z = x^2 + y^2$ and above the disk $x^2 + y^2 \leq 25$.
- Below the cone $z = \sqrt{x^2 + y^2}$ and above the ring $1 \leq x^2 + y^2 \leq 4$
- Below the plane $2x + y + z = 4$ and above the disk $x^2 + y^2 \leq 1$.
- Inside the sphere $x^2 + y^2 + z^2 = 16$ and outside the cylinder $x^2 + y^2 = 4$.
- Bounded by the paraboloid $z = 1 + 2x^2 + 2y^2$ and the plane $z = 7$ in the first octant.
- Above the cone $z = \sqrt{x^2 + y^2}$ and below the sphere $x^2 + y^2 + z^2 = 1$.
- Bounded by the paraboloids $z = 6 - x^2 - y^2$ and $z = 2x^2 + 2y^2$.
- Inside both the cylinder $x^2 + y^2 = 4$ and the ellipsoid $4x^2 + 4y^2 + z^2 = 64$.
- Above the cone $z = \sqrt{x^2 + y^2}$ and below the sphere $x^2 + y^2 + z^2 = 1$.
- Find the volume of the solid that is enclosed by the cone $z = \sqrt{x^2 + y^2}$ and the sphere $x^2 + y^2 + z^2 = 2$.
- Find the volume of the solid that lies between the paraboloid $z = x^2 + y^2$ and the sphere $x^2 + y^2 + z^2 = 2$.

- 1) Find the volume of the region E that lies between the paraboloid $z = 24 - x^2 - y^2$ and the cone $z = 2\sqrt{x^2 + y^2}$.

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Topic 11. Tripple Integral

Problem 55

a) Evaluate

$$\iiint_E \sin y \, dV,$$

where E lies below the plane $z = x$ and above the triangular region with vertices

$$(0, 0, 0), \quad (\pi, 0, 0), \quad (0, \pi, 0).$$

b) Evaluate

$$\iiint_E 6xy \, dV,$$

where E lies under the plane

$$z = 1 + x + y$$

and above the region in the xy -plane bounded by the curves

$$y = \sqrt{x}, \quad y = 0, \quad x = 1.$$

c) Evaluate

$$\iiint_E (x - y) \, dV,$$

where E is enclosed by the surfaces

$$z = x^2 - 1, \quad z = 1 - x^2, \quad y = 0, \quad y = 2.$$

d) Evaluate

$$\iiint_T y^2 \, dV,$$

where T is the solid tetrahedron with vertices

$$(0, 0, 0), \quad (2, 0, 0), \quad (0, 2, 0), \quad (0, 0, 2).$$

e) Evaluate

$$\iiint_T xz \, dV,$$

where T is the solid tetrahedron with vertices

$$(0, 0, 0), \quad (1, 0, 1), \quad (0, 1, 1), \quad (0, 0, 1).$$

f) Evaluate

$$\iiint_E x \, dV,$$

where E is bounded by the paraboloid

$$x = 4y^2 + 4z^2$$

and the plane

$$x = 4.$$

g) Evaluate

$$\iiint_E z \, dV,$$

where E is bounded by the cylinder

$$y^2 + z^2 = 9$$

and the planes

$$x = 0, \quad y = 3x, \quad z = 0$$

in the first octant.

h) Use a triple integral to find the volume of the tetrahedron enclosed by the coordinate planes and the plane

$$2x + y + z = 4.$$

i) Use a triple integral to find the volume of the solid enclosed by the paraboloids

$$y = x^2 + z^2 \quad \text{and} \quad y = 8 - x^2 - z^2.$$

j) Use a triple integral to find the volume of the solid enclosed by the cylinder

$$y = x^2$$

and the planes

$$z = 0 \quad \text{and} \quad y + z = 1.$$

k) Use a triple integral to find the volume of the solid enclosed by the cylinder

$$x^2 + z^2 = 4$$

and the planes

$$y = -1 \quad \text{and} \quad y + z = 4.$$

Problem 56 Cylindrical coordinates

a) Evaluate

$$\iiint_E \sqrt{x^2 + y^2} \, dV,$$

where E is the region that lies inside the cylinder

$$x^2 + y^2 = 16$$

and between the planes $z = -5$ and $z = 4$.

b) Evaluate

$$\iiint_E z \, dV,$$

where E is enclosed by the paraboloid

$$z = x^2 + y^2$$

and the plane $z = 4$.

c) Evaluate

$$\iiint_E (x + y + z) \, dV,$$

where E is the solid in the first octant that lies under the paraboloid

$$z = 4 - x^2 - y^2.$$

d) Evaluate

$$\iiint_E (x - y) \, dV,$$

where E is the solid that lies between the cylinders

$$x^2 + y^2 = 1 \quad \text{and} \quad x^2 + y^2 = 16,$$

above the xy -plane, and below the plane

$$z = y + 4.$$

e) Evaluate

$$\iiint_E x^2 \, dV,$$

where E is the solid that lies within the cylinder

$$x^2 + y^2 = 1,$$

above the plane $z = 0$, and below the cone

$$z^2 = 4x^2 + 4y^2.$$

f) Find the volume of the solid that lies within both the cylinder

$$x^2 + y^2 = 1$$

and the sphere

$$x^2 + y^2 + z^2 = 4.$$

g) Find the volume of the solid that is enclosed by the cone

$$z = \sqrt{x^2 + y^2}$$

and the sphere

$$x^2 + y^2 + z^2 = 2.$$

h) Find the volume of the solid that lies between the paraboloid

$$z = x^2 + y^2$$

and the sphere

$$x^2 + y^2 + z^2 = 2.$$

Problem 57 Spherical Coordinates

a) Evaluate

$$\iiint_E y^2 z^2 dV,$$

where E lies above the cone

$$\phi = \frac{\pi}{3}$$

and below the sphere

$$\rho = 1.$$

b) Evaluate

$$\iiint_E (x^2 + y^2) dV,$$

where E lies between the spheres

$$x^2 + y^2 + z^2 = 4 \quad \text{and} \quad x^2 + y^2 + z^2 = 9.$$

c) Evaluate

$$\iiint_E y^2 dV,$$

where E is the solid hemisphere

$$x^2 + y^2 + z^2 \leq 9, \quad y \geq 0.$$

d) Evaluate

$$\iiint_E x e^{x^2 + y^2 + z^2} dV,$$

where E is the portion of the unit ball

$$x^2 + y^2 + z^2 \leq 1$$

that lies in the first octant.

e) Evaluate

$$\iiint_E \sqrt{x^2 + y^2 + z^2} dV,$$

where E lies above the cone

$$z = \sqrt{x^2 + y^2}$$

and between the spheres

$$x^2 + y^2 + z^2 = 1 \quad \text{and} \quad x^2 + y^2 + z^2 = 4.$$

f) Find the volume of the part of the ball

$$\rho \leq a$$

that lies between the cones

$$\phi = \frac{\pi}{6} \quad \text{and} \quad \phi = \frac{\pi}{3}.$$

g) Find the average distance from a point in a ball of radius a to its center.

h) (a) Find the volume of the solid that lies above the cone

$$\phi = \frac{\pi}{3}$$

and below the sphere

$$\rho = 4 \cos \phi.$$

(b) Find the centroid of the solid in part (a).

i) Find the volume of the solid that lies within the sphere

$$x^2 + y^2 + z^2 = 4,$$

above the xy -plane, and below the cone

$$z = \sqrt{x^2 + y^2}.$$

 Topic 12. Line Integrals (Type 1+2)

Problem 58

Evaluate the line integral, where C is the given curve.

- $\int_C y \, ds$, $C : x = t^2, y = 2t, 0 \leq t \leq 3$.
- $\int_C (x/y) \, ds$, $C : x = t^3, y = t^4, 1 \leq t \leq 2$.
- $\int_C x \, ds$, C is the arc of the parabola $y = x^2$ from $(0, 0)$ to $(1, 1)$.
- $\int_C xy^4 \, ds$, C is the right half of the circle $x^2 + y^2 = 16$.
- $\int_C xe^y \, ds$, C is the line segment from $(2, 0)$ to $(5, 4)$.
- $\int_C (x^2y + \sin x) \, dy$, C is the arc of the parabola $y = x^2$ from $(0, 0)$ to (π, π^2) .
- $\int_C e^x \, dx$, C is the arc of the curve $x = y^3$ from $(-1, -1)$ to $(1, 1)$.
- $\int_C (x + 2y) \, dx + x^2 \, dy$, C consists of line segments from $(0, 0)$ to $(2, 1)$ and from $(2, 1)$ to $(3, 0)$.
- $\int_C x^2 \, dx + y^2 \, dy$, C consists of the arc of the circle $x^2 + y^2 = 4$ from $(2, 0)$ to $(0, 2)$ followed by the line segment from $(0, 2)$ to $(4, 3)$.
- $\int_C x^2 y \, ds$, $C : x = \cos t, y = \sin t, z = t, 0 \leq t \leq \pi/2$.

Problem 59

Evaluate the line integral.

- $\int_C x^3 \, dy - xy \, dx$ where C is the line segment from $A(0, -2)$ to $B(1, 3)$.
- $\int_C (xy - 1) \, dx + x^2 y \, dy$ where C is the line segment from $A(1, 0)$ to $B(2, 2)$.
- $\int_C xy \, dx + (y - x) \, dy$ where C is the curve from $(0, 0)$ to $(1, 1)$ of
 - the line $y = x$.
 - the parabola $y = x^2$.
 - the curve $y^2 = x$.
 - the cubic $y = x^3$.

Problem 60

Evaluate the line integral $\int_C \mathbf{F} \cdot d\mathbf{r}$, where C is given by the vector function $\mathbf{r}(t)$.

- a) $\mathbf{F}(x, y) = xy^2\mathbf{i} - x^2\mathbf{j}$, $\mathbf{r}(t) = t^3\mathbf{i} + t^2\mathbf{j}$, $0 \leq t \leq 1$.
- b) $\mathbf{F}(x, y) = e^{x-1}\mathbf{i} + xy\mathbf{j}$ and C is given by $\mathbf{r}(t) = t^2\mathbf{i} + t^3\mathbf{j}$, $0 \leq t \leq 1$.
- c) $\mathbf{F}(x, y) = (x - y)\mathbf{i} + xy\mathbf{j}$ C is the arc of the circle $x^2 + y^2 = 4$ traversed counterclockwise from $(2, 0)$ to $(0, -2)$.
- d) $\mathbf{F}(x, y) = \frac{x}{\sqrt{x^2 + y^2}}\mathbf{i} + \frac{y}{\sqrt{x^2 + y^2}}\mathbf{j}$ C is the parabola $y = 1 + x^2$ from $(-1, 2)$ to $(1, 2)$.
- e) $\mathbf{F}(x, y) = xy\mathbf{i} + x^2\mathbf{j}$ where C is the portion of $x = y^2$ from $(0, 0)$ to $(1, 1)$.

Answer: $\frac{3}{5}$.

- f) $\mathbf{F}(x, y) = (3 + 2xy^2)\mathbf{i} + 2x^2y\mathbf{j}$, and C is the arc of the hyperbola $y = 1/x$ from $(1, 1)$ to $(4, \frac{1}{4})$.

 Topic 13. Green Theorem

Problem 61

Evaluate the line integral along the given positively oriented curve.

- a) $\int_C ye^x dx + 2e^x dy$, C is the rectangle with vertices $(0, 0)$, $(3, 0)$, $(3, 4)$, and $(0, 4)$.
- b) $\int_C (x^2 + y^2) dx + (x^2 - y^2) dy$, C is the triangle with vertices $(0, 0)$, $(2, 1)$, and $(0, 1)$.
- c) $\int_C (y + e^{\sqrt{x}}) dx + (2x + \cos y^2) dy$, C is the boundary of the region enclosed by the parabolas $y = x^2$ and $x = y^2$. 
- d) $\oint_C \ln(1 + y) dx - \frac{xy}{1 + y} dy$ where C is the triangle with vertices $(0, 0)$, $(2, 0)$, and $(0, 4)$.
- e) $\int_C y^4 dx + 2xy^3 dy$, C is the ellipse $x^2 + 2y^2 = 2$.
- f) $\int_C y^3 dx - x^3 dy$, C is the circle $x^2 + y^2 = 4$.
- g) $\int_C (1 - y^3) dx + (x^3 + e^{y^2}) dy$, C is the boundary of the region between the circles $x^2 + y^2 = 4$ and $x^2 + y^2 = 9$.

Answer: -4 .

- h) $\int_C y dx + (x + y^2) dy$, C is the ellipse $4x^2 + 9y^2 = 36$ with counterclockwise orientation.
- i) $\int_C y^3 dx + x^2 dy$, C is the arc of the parabola $x = 1 - y^2$ from $(0, -1)$ to $(0, 1)$.
- j) $\int_C \mathbf{F} \cdot d\mathbf{r}$ where $\mathbf{F}(x, y) = \langle y \cos x - xy \sin x, xy + x \cos x \rangle$, C is the triangle from $(0, 0)$ to $(0, 4)$ to $(2, 0)$ to $(0, 0)$.
- k) $\int_C \mathbf{F} \cdot d\mathbf{r}$ where $\mathbf{F}(x, y) = \langle e^{-x} + y^2, e^{-y} + x^2 \rangle$ C consists of the arc of the curve $y = \cos x$ from $(-\pi/2, 0)$ to $(\pi/2, 0)$ and the line segment from $(\pi/2, 0)$ to $(-\pi/2, 0)$.
- l) $\int_C \mathbf{F} \cdot d\mathbf{r}$ where $\mathbf{F}(x, y) = \langle y - \cos y, x \sin y \rangle$ C is the circle $(x - 3)^2 + (y + 4)^2 = 4$ oriented clockwise $\mathbf{F}(x, y) = \langle \sqrt{x^2 + 1}, \tan^{-1} x \rangle$, C is the triangle from $(0, 0)$ to $(1, 1)$ to $(0, 1)$ to $(0, 0)$.
- m) $\int_C \sqrt{1 + x^3} dx + 2xy dy$ where C is the triangle with vertices $(0, 0)$, $(1, 0)$, and $(1, 3)$.
- n) $\int_C x^2 y dx - xy^2 dy$, where C is the circle $x^2 + y^2 = 4$ with counterclockwise orientation.
- o) $\int_C \mathbf{F} \cdot d\mathbf{r}$, where $\mathbf{F}(x, y) = \langle x^2 + y, 3x - y^2 \rangle$ and C is the positively oriented boundary curve of a region D that has area 6.

Topic 14. Applications of Line Integrals (Type 1+2)

Problem 62

- Find the area of one side of the fence whose base is the right half of the unit circle centered at the origin and whose height above a point (x, y) is $f(x, y) = xy^2$.

Final term 2020.
- The base of a circular fence with radius 10m is given by $x = 10 \cos t$, $y = 10 \sin t$. The height of the fence at position (x, y) is given by the function $h(x, y) = 4 + 0.01(x^2 - y^2)$, so the height varies from 3m to 5m. Suppose that 1L of paint covers 100m^2 . Sketch the fence and determine how much paint you will need if you paint both sides of the fence.
- Tom Sawyer is whitewashing a picket fence. The bases of the the fenceposts are arranged in the xy -plane as the quarter circle $x^2 + y^2 = 25$, $x, y \geq 0$, and the height of the fencepost at point (x, y) is given by $h(x, y) := 10 - x - y$ (units are feet).
 - Give a rough sketch of the fence in $\mathbb{R}^3$.
 - Use a scalar line integral to find the area of one side of the fence.

4. The base of the fence runs along a quarter-circle of radius 2. The height of the fence is given by the function $f(x, y) = 3 - x - y$. Using a scalar line integral to find the area of one side of the fence.

Problem 63

- (a) Find the work done by the force field $F(x, y) = (-x^2y + e^{\sin x}, y^2x + e^{\cos y})$ on a particle that moves once around the unit circle centered at the origin.
- (b) Find the work done by the force field $\mathbf{F}(x, y) = x^2\mathbf{i} + xy\mathbf{j}$ on a particle that moves once around the circle $x^2 + y^2 = 4$ oriented in the counterclockwise direction.
- Final term 2020.
- (c) Find the work done by the force field $\mathbf{F}(x, y) = (x^2y, xy^2)$ in moving an object along an circle $x^2 + y^2 = 1$.
- (d) Find the work done by the force field $\mathbf{F}(x, y) = x^2\mathbf{i} + ye^x\mathbf{j}$ on a particle that moves once around the unit circle centered at the origin (oriented clockwise).

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📁 Topic 15. Surface Integral

$$\iint_S f(x, y, z) dS = \iint_D f(\mathbf{r}(u, v)) |\mathbf{r}_u \times \mathbf{r}_v| dA$$

For the special case of a surface S with equation $z = f(x, y)$, where (x, y) lies in D and f has continuous partial derivatives, we take x and y as parameters. The parametric equations are

$$x = x \quad y = y \quad z = f(x, y).$$

$$A(S) = \iint_D \sqrt{1 + \left(\frac{\partial f}{\partial x}\right)^2 + \left(\frac{\partial f}{\partial y}\right)^2} dA$$

$$\iint_S \mathbf{F} \cdot d\mathbf{S} = \iint_D \mathbf{F} \cdot (\mathbf{r}_u \times \mathbf{r}_v) dA$$

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Problem 64

- a) Let a surface S be the part of the plane with vector equation $\mathbf{r}(u, v) = \langle u + v, 2 - 3u, 1 + u - v \rangle$ that is given by $0 \leq u \leq 2, -1 \leq v \leq 1$.
- Evaluate the area of one side of S .
 - Evaluate the total flux of $\mathbf{F}(x, y, z) = y\mathbf{j} - z\mathbf{k}$ through S with upward orientation.
- b) Let a surface S be the surface $z = \frac{2}{3} \left(x^{3/2} + y^{3/2} \right), 0 \leq x \leq 1, 0 \leq y \leq 1$.
- Evaluate the area of one side of S .
 - Evaluate the total flux of $\mathbf{F}(x, y, z) = y\mathbf{j} - z\mathbf{k}$ through S with upward orientation.
- c) Let a surface S be the surface with parametric equations $x = u^2, y = uv, z = \frac{1}{2}v^2, 0 \leq u \leq 1, 0 \leq v \leq 2$.
- Evaluate the area of one side of S .
 - Evaluate the total flux of $\mathbf{F}(x, y, z) = x\mathbf{i} + 2y\mathbf{j} + 3z\mathbf{k}$ through S with upward orientation.
- d) Let S be the surface with vector equation $\mathbf{r}(u, v) = \langle 2uv, u^2 - v^2, u^2 + v^2 \rangle, u^2 + v^2 \leq 1$.
- Evaluate the area of one side of S .
 - Evaluate the total flux of $\mathbf{F}(x, y, z) = y\mathbf{j} - z\mathbf{k}$ through S with upward orientation.
- e) Let S be the surface with vector equation $\mathbf{r}(u, v) = \langle u \cos v, u \sin v, v \rangle, 0 \leq u \leq 1, 0 \leq v \leq \pi$.
- Evaluate the area of one side of S .
 - Evaluate the total flux of $\mathbf{F}(x, y, z) = x\mathbf{i} + 2y\mathbf{j} + 3z\mathbf{k}$ through S with upward orientation.
- f) Let S be the cone with parametric equations $x = u \cos v, y = u \sin v, z = u, 0 \leq u \leq 1, 0 \leq v \leq \pi/2$.
- Evaluate the area of one side of S .
 - Evaluate the total flux of $\mathbf{F}(x, y, z) = y\mathbf{j} - z\mathbf{k}$ through S with upward orientation.

Problem 65

- a) $\iint_S x^2 y z \, dS$, where S is the part of the plane $z = 1 + 2x + 3y$ that lies above the rectangle $[0, 3] \times [0, 2]$.
- b) $\iint_S x z \, dS$, where S is the part of the plane $2x + 2y + z = 4$ that lies in the first octant.

- c) $\iint_S y \, dS$, where S is the surface $z = \frac{2}{3}(x^{3/2} + y^{3/2})$, $0 \leq x \leq 1, 0 \leq y \leq 1$.
- d) $\iint_S (x^2 z + y^2 z) \, dS$, where S is the hemisphere $x^2 + y^2 + z^2 = 4, z \geq 0$.
- e) $\iint_S x \, dS$, where S is the surface $y = x^2 + 4z$, $0 \leq x \leq 1, 0 \leq z \leq 1$.
- f) $\iint_S y^2 \, dS$, where S is the part of the sphere $x^2 + y^2 + z^2 = 1$ that lies above the cone $z = \sqrt{x^2 + y^2}$.
- g) $\iint_S z \, dS$, where S is the part of the paraboloid $z = x^2 + y^2$ that lies under the plane $z = 4$.
- h) $\iint_S (x^2 z + y^2 z) \, dS$, where S is the part of the plane $z = 4 + x + y$ that lies inside the cylinder $x^2 + y^2 = 4$.

Problem 66

Evaluate the surface integral $\iint_S \mathbf{F} \cdot d\mathbf{S}$ for the given vector field $\mathbf{F}$ and the oriented surface S . In other words, find the flux of $\mathbf{F}$ across S . For closed surfaces, use the positive (outward) orientation.

- a) $\mathbf{F}(x, y, z) = xy\mathbf{i} + yz\mathbf{j} + zx\mathbf{k}$, S is the part of the paraboloid $z = 4 - x^2 - y^2$ that lies above the square $0 \leq x \leq 1, 0 \leq y \leq 1$, and has upward orientation.
- b) $\mathbf{F}(x, y, z) = -x\mathbf{i} - y\mathbf{j} + z^3\mathbf{k}$, S is the part of the cone $z = \sqrt{x^2 + y^2}$ between the planes $z = 1$ and $z = 3$ with downward orientation.
- c) $\mathbf{F}(x, y, z) = x\mathbf{i} + y\mathbf{j} + z^2\mathbf{k}$, S is the sphere with radius 1 and center the origin.
- d) $\mathbf{F}(x, y, z) = y\mathbf{i} - x\mathbf{j} + 2z\mathbf{k}$, S is the hemisphere $x^2 + y^2 + z^2 = 4, z \geq 0$, oriented downward.
- e) $\mathbf{F}(x, y, z) = xz\mathbf{i} - 2y\mathbf{j} + 3x\mathbf{k}$ and S is the sphere $x^2 + y^2 + z^2 = 4$ with outward orientation.
- f) $\mathbf{F}(x, y, z) = x^2\mathbf{i} + xy\mathbf{j} + z\mathbf{k}$ and S is the part of the paraboloid $z = x^2 + y^2$ below the plane $z = 1$ with upward orientation.

Problem 67

Find the area of the surface.

- a) The part of the plane $3x + 2y + z = 6$ that lies in the first octant.
- b) The part of the plane with vector equation $\mathbf{r}(u, v) = \langle u + v, 2 - 3u, 1 + u - v \rangle$ that is given by $0 \leq u \leq 2, -1 \leq v \leq 1$.

- c) The part of the plane $x + 2y + 3z = 1$ that lies inside the cylinder $x^2 + y^2 = 3$.
- d) The part of the cone $z = \sqrt{x^2 + y^2}$ that lies between the plane $y = x$ and the cylinder $y = x^2$.
- e) The part of the surface $z = 4 - 2x^2 + y$ that lies above the triangle with vertices $(0, 0)$, $(1, 0)$, and $(1, 1)$.
- f) The part of the surface $z = xy$ that lies within the cylinder $x^2 + y^2 = 1$.
- g) The part of the surface $x = z^2 + y$ that lies between the planes $y = 0$, $y = 2$, $z = 0$, and $z = 2$.
- h) The part of the paraboloid $y = x^2 + z^2$ that lies within the cylinder $x^2 + z^2 = 16$.
- i) The helicoid (or spiral ramp) with vector equation $\mathbf{r}(u, v) = u \cos v \mathbf{i} + u \sin v \mathbf{j} + v \mathbf{k}$, $0 \leq u \leq 1$, $0 \leq v \leq \pi$.
- j) The surface with parametric equations $x = u^2$, $y = uv$, $z = \frac{1}{2}v^2$, $0 \leq u \leq 1$, $0 \leq v \leq 2$.

Problem 68

If the equation of a surface S is $z = f(x, y)$, where $x^2 + y^2 \leq R^2$, and you know that $|f_x| \leq 1$ and $|f_y| \leq 1$, what can you say about $A(S)$?

Problem 69

Use the Midpoint Rule for double integrals (see Section 15.1) with six squares to estimate the area of the surface $z = 1/(1 + x^2 + y^2)$, $0 \leq x \leq 6$, $0 \leq y \leq 4$.

Problem 70

Let E be the solid that lies below the sphere $x^2 + y^2 + z^2 = 2$ and above the paraboloid $z = x^2 + y^2$.

- a) Find the volume of E .
- b) Find the area of the boundary of E .

0.1 TRUE-FALSE QUIZ

TRUE-FALSE QUIZ

Determine whether the statement is true or false. If it is true, explain why. If it is false, explain why or give an example that disproves the statement.

a) There exists a function f with continuous second-order partial derivatives such that $f_x(x, y) = x + y^2$ and $f_y(x, y) = x - y^2$

b) $f_{xy} = \frac{\partial^2 f}{\partial x \partial y}$

c) $D_{\mathbf{k}}f(x, y, z) = f_z(x, y, z)$

d) If $f(x, y) \rightarrow L$ as $(x, y) \rightarrow (a, b)$ along every straight line through (a, b) , then $\lim_{(x,y) \rightarrow (a,b)} f(x, y) = L$.

e) If $f_x(a, b)$ and $f_y(a, b)$ both exist, then f is differentiable at (a, b) .

f) 7. If f has a local minimum at (a, b) and f is differentiable at (a, b) , then $\nabla f(a, b) = \mathbf{0}$.

g) If f is a function, then

$$\lim_{(x,y) \rightarrow (2,5)} f(x, y) = f(2, 5)$$

h) If $f(x, y) = \ln y$, then $\nabla f(x, y) = 1/y$.

i) If $(2, 1)$ is a critical point of f and

$$f_{xx}(2, 1)f_{yy}(2, 1) < [f_{xy}(2, 1)]^2$$

then f has a saddle point at $(2, 1)$.

j) If $f(x, y) = \sin x + \sin y$, then $-\sqrt{2} \leq D_{\mathbf{u}}f(x, y) \leq \sqrt{2}$.

k) If $f(x, y)$ has two local maxima, then f must have a local minimum.

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